

GitHub Actions CI/CD for Flutter

This document shows how GitHub Actions can be integrated into a Flutter application to automate tasks like testing, building, and deploying the app.

GitHub Actions Workflow File: flutter_ci.yml

Below is an example GitHub Actions YAML configuration file for setting up CI for a Flutter app.

```
name: Flutter CI

on:
  push:
    branches: [ main ]
  pull_request:
    branches: [ main ]

jobs:
  build:

    runs-on: ubuntu-latest

    steps:
      - name: Checkout Repository
        uses: actions/checkout@v3

      - name: Set up Flutter
        uses: subosito/flutter-action@v2
        with:
          flutter-version: '3.13.0'

      - name: Install Dependencies
        run: flutter pub get

      - name: Analyze Project Source
        run: flutter analyze

      - name: Run Tests
        run: flutter test
```

Step-by-step Description:

Checkout Repository pulls your code from GitHub so the workflow can use it.

Set up Flutter installs the specified version of Flutter so commands like flutter pub get and flutter test work.

Install Dependencies downloads the packages listed in your pubspec.yaml

Analyse Project Source runs flutter analyse to check for potential issues inside of the code.

Run Tests Executes all the unit tests in your project using flutter test